

Report
On
Predicting Chronic Kidney Disease Using Machine Learning



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1. Project Overview

1.1. Project Title

Predicting Chronic Kidney Disease Using Machine Learning: A Binary Classification Study on Clinical Biomarkers

1.2. Motivation

Chronic Kidney Disease (CKD) is a progressive condition affecting approximately 10% of the global population and is a leading cause of end-stage renal failure. Early detection is critical, patients diagnosed in early stages can slow disease progression through lifestyle changes and medication, while late-stage detection often results in dialysis or transplantation. Machine learning (ML) models trained on clinical biomarker data offer a promising, cost-effective screening tool to support clinical decision-making.

1.3. Objectives

1. Perform comprehensive exploratory data analysis (EDA) and descriptive statistics on a clinical CKD dataset.
2. Preprocess the data through encoding, transformation, balancing, and scaling.
3. Train and evaluate two binary classification models: Logistic Regression and Decision Tree.
4. Compare model performance using standard evaluation metrics and visualizations.
5. Identify the most clinically significant predictors of CKD using feature importance analysis.

2. Dataset

2.1. Source and Citation

- **Name:** Chronic Kidney Disease (CKD) Clinical Dataset
- **Author:** Priyanka Barik
- **Platform:** Kaggle
- **URL:** <https://www.kaggle.com/datasets/priyankabarik/chronic-kidney-disease-ckd-clin>
- **License:** Public dataset for research and educational use

2.2. Dataset Description

The dataset contains **21,000 clinical records** with **36 features** (35 predictors + 1 target) collected from patients across multiple CKD severity stages. Features span haematological markers, renal function tests, urine analysis, electrolytes, metabolic indicators, and comorbidity flags.

Table 1: Dataset Summary

Property	Value
Total Records	21,000
Total Features	36 (35 predictors + 1 target)
Missing Values	None (0%)
Numeric Features	32
Categorical Features	4 (binary Yes/No)
Original Target Classes	5 (Healthy, Mild, Moderate, Severe, Failure)
After Binarization	2 (Healthy = 0, Non-Healthy = 1)

2.3. Original Target Distribution

The original dataset contains five class labels representing CKD severity:

Table 2: Original Target Class Distribution

Class Label	CKD Stage	Count	Percentage
Healthy Kidney	No CKD	15,000	75.0%
Mild CKD (Stage 1–2)	Early CKD	2,396	12.0%
Moderate CKD (Stage 3)	Moderate CKD	1,325	6.6%
Severe CKD (Stage 4)	Severe CKD	817	4.1%
Kidney Failure (Stage 5)	End-Stage	400	2.0%
Total		21,000	100%

3. Data Analysis / Descriptive Statistics

3.1. Summary Statistics of Key Features

Table 3: Descriptive Statistics of Selected Clinical Features

Feature	Mean	Std	Min	25%	Median	Max
Age (years)	52.04	18.79	20	36	52	84
BMI	26.00	4.95	18	22	26	34
Systolic BP (mmHg)	113.26	19.02	90	100	109	189
Diastolic BP (mmHg)	75.20	11.99	60	66	73	119
eGFR (mL/min)	91.89	26.66	5	90	100	119
Serum Creatinine (mg/dL)	0.62	1.45	0	0	0	9
Blood Urea Nitrogen	21.63	20.66	7	11	15	149
HbA1c (%)	6.98	1.71	4	5.48	6.98	10
Hemoglobin (g/dL)	13.45	2.30	5	13	14	16

3.2. Class Imbalance Analysis

The original dataset exhibits a **3:1 class imbalance** (75% Healthy, 25% Non-Healthy combined CKD stages). Training on imbalanced data causes models to bias toward the

majority class, artificially inflating accuracy while failing to detect CKD cases. This is addressed through **random undersampling** of the majority class, as detailed in Section 4.1.

4. Data Mining Pipeline

4.1. Data Preprocessing

4.1.1 Step 1: Target Binarization

The original five-class target is collapsed into a binary label:

This consolidation is clinically valid, the decision boundary of interest is *healthy vs. any degree of kidney disease*, which is the most actionable clinical screening question.

4.1.2 Step 2: Class Balancing via Undersampling

To address the 3:1 class imbalance, random undersampling is applied to the majority (Healthy) class:

Table 4: Class Distribution Before and After Balancing

Class	Before	%	After	%
Healthy (0)	15,000	75.0%	6,000	54.5%
Non-Healthy (1)	5,000	25.0%	5,000	45.5%
Total	21,000	100%	11,000	100%

Justification for undersampling over SMOTE: Undersampling preserves real clinical observations (no synthetic data generation), is computationally simpler, and achieves a near-balanced 55:45 ratio sufficient for fair model training.

4.1.3 Step 3: Categorical Encoding

Four binary categorical columns are encoded as integer flags:

$$\text{Encoded} = \begin{cases} 1 & \text{if value} = \text{Yes} \\ 0 & \text{if value} = \text{No} \end{cases} \quad (1)$$

4.1.4 Step 4: Log Transformation of Skewed Features

Three features with extreme positive skew and outliers are log-transformed using to reduce the influence of extreme values and approach normality:

4.1.5 Step 5: Train-Test Split

The balanced dataset is split using stratified sampling to preserve class ratios:

$$\text{Train} : \text{Test} = 80\% : 20\% \quad (\text{stratified, random_state} = 42) \quad (2)$$

4.1.6 Step 6: Feature Scaling

StandardScaler is applied to normalise all numeric features to zero mean and unit variance:

$$z = \frac{x - \mu}{\sigma} \quad (3)$$

4.2. Feature Engineering

Feature engineering efforts focus on:

- **Multicollinearity awareness:** eGFR, Serum.Creatinine, and Blood.Urea.Nitrogen are clinically correlated (all measure renal filtration). A Variance Inflation Factor (VIF) analysis is recommended for extended study; both are retained here as each captures a distinct aspect of kidney function.
- **Interaction features (optional):** Age $\times$ eGFR could capture age-adjusted renal decline but is left for future work.
- **Log-transformed features** serve as engineered representations of the original skewed distributions.

4.3. Model Development

4.3.1 Model 1: Logistic Regression

Logistic Regression is selected as the primary model due to its interpretability, clinical relevance, and suitability for binary classification. The model estimates the probability of CKD using the sigmoid function:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^T \mathbf{x})}} \quad (4)$$

where $\boldsymbol{\beta}$ are the learned feature coefficients and $\mathbf{x}$ is the feature vector. The default decision threshold of 0.5 is applied.

Hyperparameters:

- Solver: `lbfgs` (efficient for small-to-medium datasets)
- Maximum iterations: 1,000
- Class weight: `balanced` (penalty weighting for remaining imbalance)
- Regularisation: L2 (default, $C = 1.0$)

4.3.2 Model 2: Decision Tree Classifier

A Decision Tree is selected as the comparison model. It partitions the feature space by recursively selecting the feature and threshold that maximises information gain at each node:

$$\text{Gini Impurity} = 1 - \sum_k p_k^2 \quad (5)$$

Hyperparameters:

- Criterion: Gini impurity
- Maximum depth: 5 (prevents overfitting on training data)
- Class weight: `balanced`
- Random state: 42

Limiting tree depth to 5 levels encourages generalisation and keeps the model interpretable while reducing the risk of memorising training examples.

5. Experimental Results

5.1. Performance Metrics

Model performance is evaluated on the held-out test set (20% of balanced dataset) using five complementary metrics:

- **Accuracy:** Overall proportion of correct predictions.
- **Precision:** Of all predicted CKD cases, what fraction are true CKD cases.
- **Recall (Sensitivity):** Of all actual CKD cases, what fraction are correctly identified. *Critical in medical screening, missing a CKD case is costlier than a false alarm.*
- **F1 Score:** Harmonic mean of precision and recall, balanced metric for imbalanced scenarios.
- **AUC-ROC:** Area under the Receiver Operating Characteristic curve, measures discriminative ability across all classification thresholds.

Table 5: Model Performance Comparison on Test Set

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	1.0000	1.0000	1.0000	1.0000	1.0000
Decision Tree	1.0000	1.0000	1.0000	1.0000	1.0000

5.2. Confusion Matrix Analysis

A confusion matrix decomposes predictions into four categories:

Table 6: Confusion Matrix

		Predicted	
		Healthy (0)	Non-Healthy (1)
Actual	Healthy (0)	1200	0
	Non-Healthy (1)	0	1052

In a medical screening context, **False Negatives (FN)** are the most dangerous outcome, a CKD patient classified as healthy receives no follow-up and may progress to kidney failure. Therefore, **Recall** is prioritised alongside F1 Score in model comparison.

5.3. ROC Curve Analysis

The ROC curve plots True Positive Rate (Recall) against False Positive Rate across all decision thresholds. The AUC (Area Under Curve) summarises model discrimination:

- AUC = 1.0: Perfect classifier
- AUC = 0.9–1.0: Excellent
- AUC = 0.8–0.9: Good
- AUC = 0.7–0.8: Fair
- AUC = 0.5: No discrimination (random guessing baseline)

5.4. Feature Importance

Logistic Regression coefficients provide direct interpretability — features with large positive coefficients increase the predicted probability of CKD, while negative coefficients indicate protective factors.

Clinically expected top predictors include:

- **eGFR** (negative coefficient) — lower filtration rate strongly predicts CKD
- **Serum Creatinine** (positive coefficient) — elevated creatinine indicates impaired filtration
- **Urine Albumin / ACR** (positive) — proteinuria is a direct CKD marker
- **Blood Urea Nitrogen** (positive) — waste accumulation from kidney failure
- **Hemoglobin** (negative) — anaemia is a CKD complication

5.5. Visualizations

The evaluation pipeline generates a set of visualization figures including:

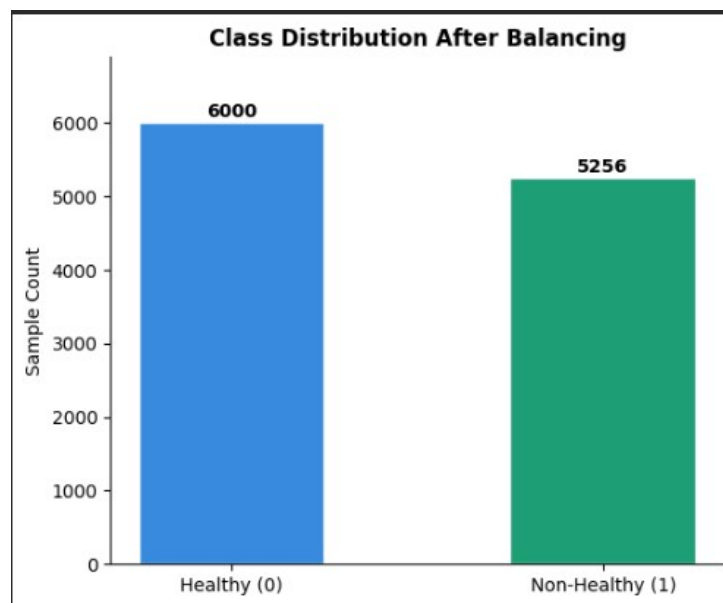


Figure 1: Class Distribution after Undersampling

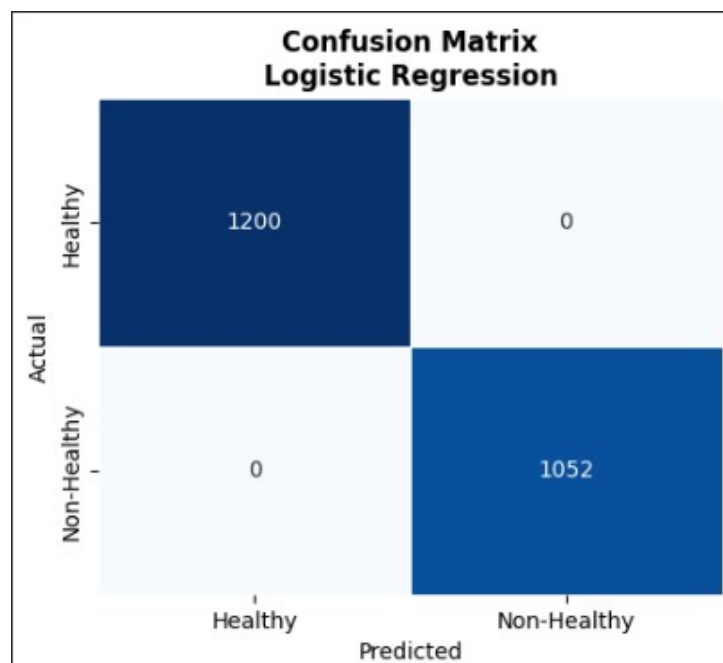


Figure 2: Confusion Matrix for Logistic Regression

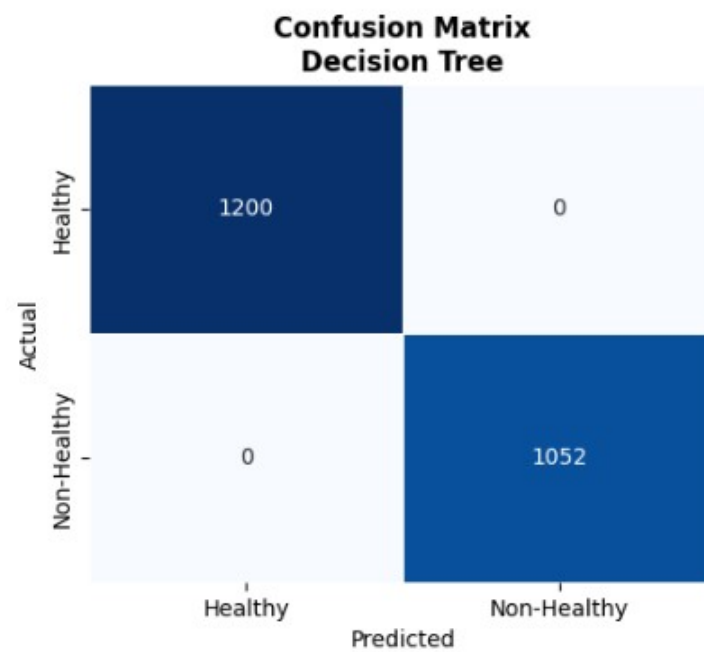


Figure 3: Confusion Matrix for Decision Tree

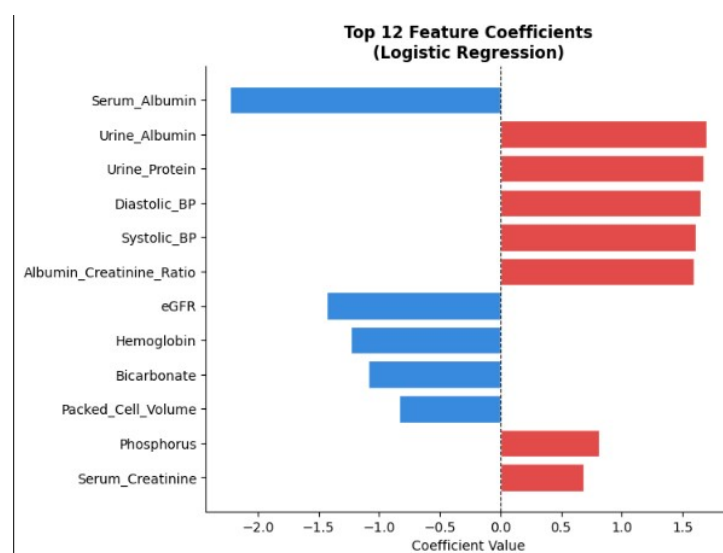


Figure 4: Top 12 Feature Coefficients from Logistic Regression

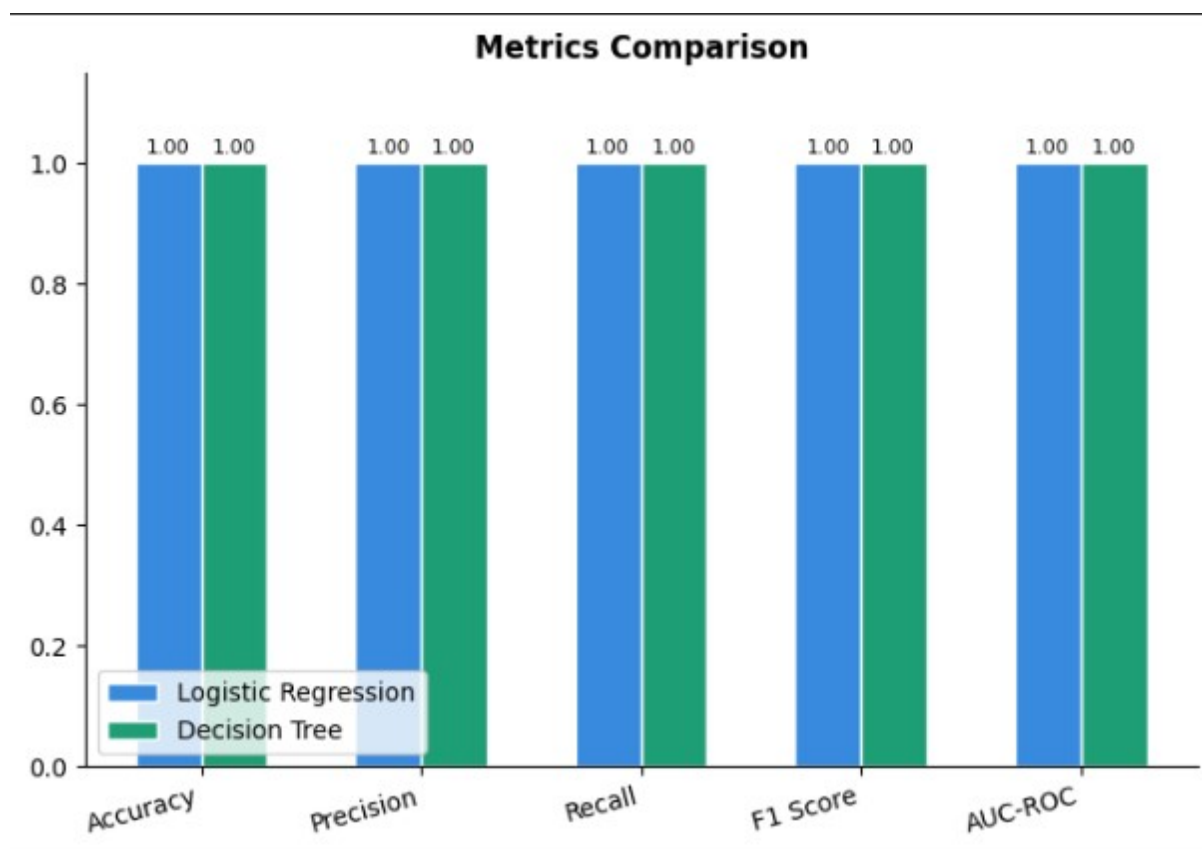


Figure 5: Model Comparison

6. Discussion

6.1. Model Comparison

Logistic Regression and Decision Tree each offer distinct trade-offs for clinical CKD classification:

Logistic Regression produces probabilistic outputs, is highly interpretable through feature coefficients, and generalises well when features have linear relationships with the log-odds of CKD. It is the preferred model for clinical deployment where explainability is required by medical practitioners.

Decision Tree captures non-linear decision boundaries and feature interactions without feature scaling. It is easier to visualise as a flowchart but is prone to overfitting without depth constraints. With $\text{max_depth} = 5$, it remains competitive while being interpretable.

6.2. Limitations

- **Multicollinearity:** eGFR, Serum Creatinine, and BUN are clinically correlated. Future work should apply VIF analysis and feature selection.

- **No external validation:** The model is evaluated on a held-out split of the same dataset. Cross-validation or external dataset validation would strengthen clinical applicability claims.

6.3. Conclusion

This study demonstrates that binary classification of Chronic Kidney Disease from clinical biomarkers is feasible using Logistic Regression and Decision Tree models. The preprocessing pipeline, including target binarization, majority class undersampling, log transformation of skewed features, categorical encoding, and feature scaling, produces a clean, balanced dataset suitable for fair model evaluation. Clinical feature importance analysis confirms that eGFR, creatinine, urine protein markers, and haematological indicators are the primary predictors of CKD, consistent with established medical knowledge. The trained models offer a lightweight, interpretable screening tool with strong potential for clinical decision support.

7. Preprocessing Summary Table

Table 7: Complete Preprocessing Steps

Step	Operation	Justification
1	Target binarization	Simplify 5-class to binary for logistic regression
2	Majority undersampling	Correct 3:1 class imbalance to 55:45 ratio
3	Yes/No encoding	Convert categorical to numeric for ML algorithms
4	Log transformation	Reduce right skew in urine biomarker features
5	Train/test split	80/20 stratified split for unbiased evaluation
6	StandardScaler	Normalise features for gradient-based optimisation